Reliability-Oriented K-Means in StatCore: Exact Variants, Objectives, and Empirical Limits

Jabulani Monchwe
Applied Business Analytics and Decision Making
North-West University
IDEA-Research Club Chairperson
Johannesburg, South Africa
monchwej.25@gmail.com

Martin Chanza
Applied Business Analytics and Decision Making
North-West University
Mafikeng, South Africa
martin.chanza@nwu.ac.za

Olorato Motitswane
Eduvos Midrand Campus – IT Faculty Data Science, South Africa
IDEA-Research Club
oloratomotitswane@gmail.com

Mfundo Monchwe
Eduvos Midrand Campus – IT Faculty Data Science, South Africa
IDEA-Research Club
diditmfundo@gmail.com

***Abstract*—This paper gives an implementation-exact account of the pure-NumPy StatCore K-means family. We distinguish Lloyd's squared-Euclidean algorithm with random starts from k-means++-seeded Lloyd optimization; specify graph-smoothed assignment, trimmed coordinatewise k-medians, and BIC/silhouette cluster-count selection; and derive the standard objective and block updates. The topology-aware update is not descent on ordinary inertia, and the robust objective differs from the public squared-Euclidean inertia. We state convergence and computational limits without assigning textbook guarantees to modified methods. Repeated-seed experiments on contaminated synthetic clusters and bundled Iris compare every variant to scikit-learn K-means using external adjusted Rand index, silhouette, and squared-Euclidean inertia. Results are measured and include the possibility of regressions, not evidence of universal superiority.**

***Index Terms*—K-means, clustering, robust statistics, graph regularization, model selection, reliability**

I. Introduction

K-means minimizes within-cluster squared Euclidean distance for a fixed number of clusters. Lloyd's alternating assignment and centroid updates are simple but depend on initialization and can terminate at local minima [1], [2]. K-means++ samples seeds proportional to squared distance and has an expected approximation guarantee for its initial potential, not a global guarantee for a finite Lloyd run [3]. StatCore adds topology, robust, and model-selection variants to a standard baseline and an initialization variant. This paper documents `src/ml_reliability/statcore/kmeans/kmeans.py`; method names do not imply stronger theory than the code warrants. Inputs are finite numeric $X \in \mathbb{R}^{n\times d}$, at least two rows and one feature. Fixed k is integer in $[1, n]$. Optional labels passed to `fit` are ignored. Defaults: $k = 3$, `max_iter`=100, `n_init`=5, `tol`=10^{-6}, `random_state`=0.

II. Standard K-means and Lloyd's algorithm

For assignments $z_i \in \{1, \dots, k\}$ and centers μ_j,

$$J(z, \mu) = \sum_i \|x_i - \mu_{z_i}\|_2^2. \tag{1}$$

For fixed centers, assigning each row to the nearest center minimizes J independently. For fixed nonempty cluster S_j, differentiation gives $\nabla_{\mu_j} \sum_{i\in S_j} \|x_i - \mu_j\|^2 = 2|S_j|\mu_j - 2\sum_{i\in S_j} x_i$; setting it to zero yields the arithmetic mean, the unique minimizer. Therefore exact assignment and mean steps do not increase J. Since there are finitely many labelings, deterministic Lloyd iterations reach a fixed labeling (allowing ties and a consistent tie rule), but only a local solution. A finite iteration cap may stop early.

A. base: random row starts and best restart

Each of `n_init` runs samples k distinct rows uniformly, then alternates squared-distance assignments and arithmetic means. It keeps the run with smallest final assigned squared-distance sum. It stops only if center shift $\sum_j \|\mu_j^{new} - \mu_j\|_2^2 \le$ `tol` and labels equal the previous iteration, or at `max_iter`. Empty clusters are reseeded to the currently worst-fit row. Ties use NumPy first-index `argmin`. At most I iterations cost $O(Inkd)$ per restart with $O(nk)$ cost storage, plus initialization; best of five restarts by default. **Algorithm 1: StatCore base Lloyd run (one restart).**

1) Draw k distinct data rows as centers; set previous labels to unset.

2) Assign every row to its nearest squared-Euclidean center (first index breaks ties).
3) For each nonempty cluster, replace its center by its arithmetic mean; reseed empty clusters at the worst-fit row.
4) Stop only when squared center shift is at most `tol` and labels are unchanged, or after `max_iter`; otherwise repeat Step 2.
5) Recompute ordinary assigned SSE. Across `n_init` restarts, retain the run with the lowest SSE.

B. better_init: k-means++

The first center is a uniform random row. For each next center, with D_i^2 the minimum squared distance to already chosen centers, sample row i with probability $D_i^2 / \sum_r D_r^2$; if all distances are zero, choose uniformly. It then runs the same Lloyd and restart-selection routine as base. The classical expected $O(\log k)$ guarantee is about the initial potential under the standard objective and does not guarantee a global optimum after finite Lloyd iterations. Duplicate points are supported by the zero-potential fallback. Seeding costs $O(nkd)$ per run.

III. Modified methods

A. topology_aware: nearest-neighbor reward in assignment

A directed training kNN graph is constructed from Euclidean distances after setting diagonal distances to infinity; stable sorting selects $m = \min(\texttt{n_neighbors}, n-1)$ neighbors. The default is 10. Starts are k-means++. On first Lloyd iteration assignment is standard. Subsequently, for previous labels define q_{ij} as the fraction of row i's neighbors labeled j, and $d_{ij}^2 = \|x_i - \mu_j\|^2$. With $\bar{d}^2 = (nk)^{-1} \sum_{i,j} d_{ij}^2$, code uses

$$z_i = \arg\min_j \left[\frac{d_{ij}^2}{\bar{d}^2 + 10^{-12}} - \lambda q_{ij} \right], \qquad \lambda = \texttt{topology_weight}. \tag{2}$$

The default $\lambda = 0.5$; then ordinary unweighted means update centers. The normalization is across all point-center pairs, not rowwise. The graph term is a reward for previous local agreement, not a globally minimized graph-cut penalty. Means need not minimize the modified assignment score; ordinary SSE can increase, and no descent guarantee follows. The hard shift-and-label stopping condition is unchanged. Importantly, final labels and restart scores are recomputed using ordinary nearest-center distances; graph smoothing is not applied at finalization. Thus the selected restart minimizes final ordinary SSE. Pairwise graph construction costs $O(n^2 d + n^2 \log n)$ time and $O(n^2)$ memory; each iterative pass is approximately $O(nmk + nkd)$.

B. robust: trimmed coordinatewise k-medians

This branch assigns by Manhattan cost $a_i = \|x_i - \mu_{z_i}\|_1$, initializes uniformly without replacement, and updates each coordinate to the median of assigned points. For untrimmed assignments, median minimizes the sum of absolute deviations coordinatewise, giving block minimization of $\sum_i \|x_i - \mu_{z_i}\|_1$. With default trim fraction $\tau = 0.05$, each update retains points satisfying $a_i \le Q_{1-\tau}(a)$, NumPy's global quantile; if a cluster has no retained point, it falls back to all points assigned to it. Empty clusters are reseeded at the worst-fit point. Reassignment occurs on all rows each round; center-shift and label-stability stopping, cap, and restart strategy otherwise follow the base routine (random starts, not k-means++). With no trimming, alternating exact L1 assignment and median updates has nonincreasing L1 objective; with deterministic tie handling its finite assignment states yield a local fixed point absent premature cap. Trimming couples the retained set globally and does not provide a single fixed objective with guaranteed descent. The implementation does not validate $0 \le \tau < 1$, so invalid values may cause quantile errors or unexpected behavior. `objective_` reports final L1 costs, but `inertia_` is always squared-L2 SSE, as are shared clustering metrics. `transform` alone returns L1 for robust and Euclidean distances otherwise. These are different quantities and must not be conflated.

C. stat_model_selection: BIC or silhouette

For $\texttt{k_range} = (\ell, h)$, default $(2, 8)$, it checks $\ell \ge 2, h \ge \ell$, caps h at $n-1$, and fits every candidate with k-means++ (seed `random_state`+k), five restarts by default. If the cap makes $h < \ell$, no candidates are fit and the current implementation fails rather than falling back. Criterion does not change the candidate-fitting objective. For BIC, let SSE be ordinary final squared error, $\hat{v} = \max(\mathrm{SSE}/[\max(n-k, 1)d], 10^{-12})$, and n_j be nonzero cluster sizes. The implemented approximate log score and parameter count are

$$\ell_k = \sum_{j:n_j>0} n_j \log(n_j/n) - \tfrac{1}{2} nd \log(2\pi\hat{v}) - \tfrac{1}{2}(n-k)d,$$
$$p_k = (k-1) + kd + 1, \quad \mathrm{BIC}_k = -2\ell_k + p_k \log n.$$

The selected model has minimum BIC; `scores_[k]` stores BIC itself. This resembles a spherical common-variance hard-assignment Gaussian-mixture approximation, not the exact maximized general mixture likelihood: variance denominator is coded as above, cluster weights enter separately, and empty centers have no explicit likelihood model. BIC selection is criterion optimization over a finite range, not recovery guarantee of a true cluster count. For silhouette, pairwise Euclidean distances yield own-cluster mean a_i excluding self (denominator $n_{z_i} - 1$) and nearest-other-cluster mean b_i. Score $(b_i - a_i)/\max(a_i, b_i, 10^{-12})$ is averaged; singleton rows score zero. Fewer than two unique clusters gives zero. The maximum-silhouette candidate wins. This costs $O(n^2 d + n^2)$ time and $O(n^2)$ memory. The selection procedure does not nest validation, and a criterion optimized on the same data can be optimistic.

D. Shared API and numerical behavior

Squared distances use $\|x\|^2 + \|\mu\|^2 - 2x^\top \mu$, clipped below at zero. Inputs are finite and feature count fixed at prediction. No automatic scaling or missing-data treatment is provided;

large-scale features dominate. `predict` returns nearest-center labels (L1 only for robust), `fit_predict` training labels, and `evaluate` shared clustering metrics. None quantify assignment uncertainty or seed stability. Empty-cluster reseeding and deterministic first-index tie resolution affect trajectories.

IV. Empirical study

The attached `experiments/kmeans_experiment.py` compares each variant against scikit-learn KMeans as standard reference [6], on synthetic and bundled Iris data. Synthetic data comprise three Gaussian clusters in four dimensions and 18 broad training-only contaminants; held-out points are uncontaminated. Iris is split per seed, and its labels provide external evaluation only; they are not given to the clustering model. Training means and standard deviations standardize both train and test, replacing zero scales by one. Five seeds are 11, 23, 37, 51, 71. Reference and fixed-k methods use $k = 3$, five restarts; model selection uses BIC over 2–6. Held-out metrics are ARI versus generating/class labels (higher), silhouette (higher), and squared-L2 SSE (lower). For model selection, different k can change SSE mechanically, so inertia is not a standalone ranking. The script computes metrics directly and writes LaTeX cells only from executed output. These small benchmarks cannot prove population superiority. Synthetic contamination is controlled; Iris is small and its semantic classes need not be spherical clusters. ARI measures agreement with labels, not necessarily operational utility. Selection may prefer another k and reduce ARI. Five seeds describe limited split/init variability, not uncertainty intervals. The limitations and variants that regress should inform model choice, not be hidden.

V. Limitations and conclusion

Scaling, local minima, and distribution shift remain material. Topology smoothing can propagate wrong neighbors and is not descent; trimming can delete real rare clusters; the coded BIC is an approximation. Neither inertia nor silhouette is uncertainty calibration. Base and better initialization retain ordinary Lloyd descent; other variants modify its objectives and cannot inherit those guarantees. Reproducible, task-specific evaluation remains necessary.

References

[1] S. P. Lloyd, "Least squares quantization in PCM," *IEEE Transactions on Information Theory*, vol. 28, no. 2, pp. 129–137, 1982.

[2] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. Fifth Berkeley Symp. Mathematical Statistics and Probability*, vol. 1, 1967, pp. 281–297.

[3] D. Arthur and S. Vassilvitskii, "k-means++: The advantages of careful seeding," in *Proc. 18th Annual ACM-SIAM Symp. Discrete Algorithms*, 2007, pp. 1027–1035.

[4] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.

[5] G. Schwarz, "Estimating the dimension of a model," *The Annals of Statistics*, vol. 6, no. 2, pp. 461–464, 1978.

[6] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[7] J. Monchwe *et al.*, "ReliableML: Reliability-oriented machine learning," software repository, https://github.com/Jabualani/ML-Reliabilty, accessed Oct. 2026.

TABLE I

EXACT DIFFERENCES AMONG KMEANS VARIANTS; COMPLEXITIES ARE PER RESTART UNLESS NOTED.

Variant	Initialization and assignment	Update / selection	Objective reported	Guarantee	Defaults
base	Uniform unique rows; squared L2	Mean; best ordinary SSE	SSE	Lloyd descent to local solution (unless capped)	$k = 3, R = 5, I = 100$
better_init	D-squared k-means++; squared L2	Same as base	SSE	Same Lloyd guarantee; seeding bound is initial-potential only	seed 0
topology aware	k-means++; normalized SSE less neighbor reward	Means; final restart chosen by ordinary SSE	Transient smoothed score; public SSE	No ordinary-SSE descent guarantee	$m = 10, \lambda = .5$
robust	Uniform rows; L1 assignment	Coordinate median on retained global quantile	L1 objective; L2 inertia	Untrimmed L1 block descent; trim has no shared monotone objective	$\tau = .05$
stat. model selection	k-means++ for each candidate	Minimum approximate BIC or max silhouette	Chosen criterion plus L2 inertia	Criterion selection only; no true-k guarantee	$k = 2 \dots 8$, BIC

TABLE II

HELD-OUT METRICS, MEAN $\pm$ SAMPLE SD OVER FIVE SEEDS. HIGHER ARI/SILHOUETTE IS PREFERRED; LOWER SSE IS PREFERRED ONLY AT COMPARABLE CLUSTER COUNT.

Dataset	Method	ARI	Silhouette	SSE
Synthetic	Reference KMeans	1.0000 ± 0.0000	0.7294 ± 0.0163	15.9602 ± 1.8563
Synthetic	base	1.0000 ± 0.0000	0.7294 ± 0.0163	15.9602 ± 1.8563
Synthetic	topology_aware	0.9276 ± 0.1619	0.7052 ± 0.0658	22.7294 ± 15.9821
Synthetic	stat_model_selection	1.0000 ± 0.0000	0.7294 ± 0.0163	13.4264 ± 2.7771
Synthetic	robust	1.0000 ± 0.0000	0.7294 ± 0.0163	13.4174 ± 2.7072
Synthetic	better_init	0.9276 ± 0.1619	0.7052 ± 0.0658	22.9262 ± 15.8712
Iris	Reference KMeans	0.6017 ± 0.0971	0.4650 ± 0.0555	35.0838 ± 7.4583
Iris	base	0.6044 ± 0.0801	0.4676 ± 0.0554	34.8722 ± 7.1741
Iris	topology_aware	0.5933 ± 0.1061	0.4657 ± 0.0550	35.5854 ± 6.8826
Iris	stat_model_selection	0.4124 ± 0.1056	0.2656 ± 0.0708	23.1849 ± 4.1757
Iris	robust	0.6723 ± 0.1213	0.4597 ± 0.0319	36.5280 ± 7.4923
Iris	better_init	0.6017 ± 0.0971	0.4650 ± 0.0555	35.0838 ± 7.4583